



SpiderRock V7 Tool Concept Guide

Risk Manager

July 3, 2023

RISK MANAGER

The SpiderRock Risk Manager is a live portfolio risk management tool allowing users to analyze and quickly shape portfolio risk. Users can create custom risk slides to understand the risk sensitivity of a portfolio and address areas of concern with risk-reduction capabilities. The tool allows you to view risk across multiple accounts and send mass basket imports of orders.

The **Risk Manager** also includes a live market-scanning application designed to identify potential risk-opening opportunities based on rich user-defined filter settings.

SPIDERROCK
PLATFORM

SVOMOMCTMMMW

TRADING

RMRVHT

RISK

TA RADCD

HISTORICAL

CVSM AVFX

CONTROL

CLOSE ALL TOOLS

LAUNCH SAVED LAYOUT

SAVE CURRENT LAYOUT AS DEFAULT

RISK MANAGER

- Aggregate live equity, option, and future risk, P&L, and analytics in a pivot table construct
- Compose and analyze custom portfolio-wide risk slide scenarios
- Scan the market for potential risk-opening and risk-closing trades using rich client-defined filter macros

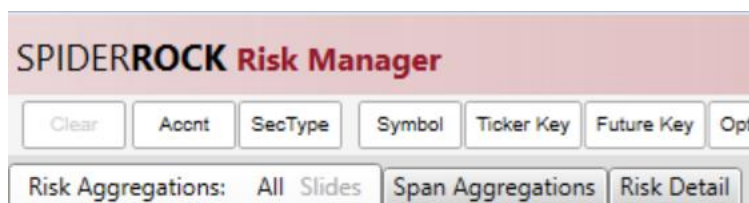
SYSTEM INFORMATION

User: pat.read (SRAdmin)
Version: v7.0 (Built: 12/18/2021 6:42 PM)
Citrix Host: SPIDERNJ131
Tool Server: 10.37.201.196:5240
Environment: V7_Stable.Prod | NY4
Time: 02:52:16.0 PM



GENERAL TOOL LAYOUT

The tool is composed of 3 different tabs has 2 general aggregation layers and additional filter buckets across the top.



The Risk Aggregations tab made up of 2 modes (**All** and **Slides**) which users can toggle between:

- **All:** Risk aggregates across a variety of measures for all asset classes composing the portfolio. This tool allows users to view both positional risk and PNL by account, symbol, expiration, etc.
- **Slides:** Clients can choose and arrange viewable slide columns from 18 customizable Risk Scenarios created in the "Scenarios" widget. Once created, clients can right-click the column headers to select the Scenarios they would like to incorporate in their Risk Slides. The cells will populate with PNL only unless user expands to view associated greeks at each slide interval (PNL, Delta, \$ Delta, Vega, WTVega, Gamma, \$Gamma and Theta). Risk Aggregation tab views can be saved and automatically categorized for All or Slide mode.

Note: The Risk Aggregations tab is organized as a split window. The **Aggregated Risk** (top) pane shows the aggregate risk, and the **Constituent Risk** (bottom) pane displays a more granular view of the top pane's risk.

The Span Aggregations tab provides a Standard Portfolio Analysis of Risk (SPAN) methodology evaluate risk under a broad range of 'what-if' market scenarios. The customary 16 slides based on an underlier price and vol change can be displayed following the CME exchange guidelines or can be modified with user input, based on price/vol settings.

The Risk Detail tab provides all risk metrics available at the series level for a selected account(s) in a non-aggregated capacity. Clients can easily export position details at the series level into Excel



RISK AGGREGATIONS TAB

- Risk Aggregations - All

The Risk Aggregations tab allows a visualization of current portfolio risk (positions plus day trades) in multiple aggregation buckets. Two aggregation lists are available across the top which enables any portfolio manager to analyze and monitor risk across accounts, symbol, industry, expiration etc.

SPIDERROCK Risk Manager (v7.12.99) [V7_Stable_Prod] [CTX/Est/Est] [© 2023 SpiderRock Technology, LLC (conan.brad@spiderrock.com)]

Agg

Account

Symbol

Blotter

Blotter (Mkt)

Scenarios

Macros

Set Binary

Default View

Test Account

TCB

Live

Risk Aggregations: All Slides

Risk Aggregations: All Slides

Account	Symbol	Expiration	SecType	Strike	Expiration	ExpCode	Shew	EDays	Ind	Sub	Grp	Sector	Undr Move Rank	Vol Move Rank	Vol Range	ViewGroup1	ViewGroup2	ViewGroup3	Total Move Rank	TEdge Move Rank	TEdge Rank
1	T.CB	AAR	2023-08-18	Call	179.00	186.9500	0.740	0.1879	691.314	8,181	0.62	0	50	50	66,493	0	0	44	0	0	0
2	T.CB	IWM	2023-08-18	Call	168.00	186.9500	0.808	0.1879	860,520	10,184	0.62	0	57	57	140,270	0	0	54	0	0	0
3	T.CB	IWM	2023-08-18	Call	178.00	186.9500	0.758	0.1879	380,332	4,525	0.62	0	27	27	34,769	0	0	24	0	0	0
4	T.CB	IWM	2023-11-17	Call	171.00	186.9500	0.781	0.1879	802,631	9,499	0.62	0	55	55	123,864	0	0	51	0	0	0
5	T.CB	IWM	2023-06-16	Put	205.00	186.9500	-1.000	0.1879	2,803,425	33,178	0.62	0	0	0	-150	-150	-196,635	0	0	178	0
6	T.CB	IWM	2023-06-16	Put	205.00	186.9500	-1.000	0.1879	2,803,425	33,178	0.62	0	0	0	-17,865	-1,260,378	-46,891	39	-1,399	2,091	0
7	T.CB	IWM	2023-12-15	Put	165.00	186.9500	-0.181	0.1879	3,382	40	0.62	0	0	0	6,810	302,123	5,211	7	-327	4,474	13,163
8	T.CB	IWM	2023-11-17	Call	170.00	186.9500	0.790	0.1879	826,994	9,787	0.62	0	56	56	126,822	0	0	52	0	0	0
9	T.CB	IWM	2023-06-16	Put	210.00	186.9500	-1.000	0.1879	2,747,356	32,514	0.62	0	0	0	-147	-147	-116,444	0	0	174	0
10	T.CB	IWM	2023-12-15	Put	240.00	186.9500	-1.000	0.1879	446,548	5,308	0.62	0	0	0	-24	-24	-127,452	0	0	28	0
11	T.CB	IWM	2023-06-16	Put	180.00	186.9500	-0.025	0.1879	3,783	45	0.62	0	0	0	-8	-8	-21	0	0	0	0
12	T.CB	IWM	2023-11-17	Call	167.00	186.9500	0.816	0.1879	869,371	10,289	0.62	0	57	57	146,194	0	0	55	0	0	0

Risk Aggregations: All Slides

Aggregated Risk Pane

Constituent Risk Pane

Connected Account:TCB

Sum: 0 Cnt: 1 Min: 0 Q1: 0 Med: 0 Q3: 0 Max: Avg: 10.37.200.160:5240/v07.12.99

Rows: 268

The tab launches on a grid (Aggregated Risk Pane) which is a default view of the client's risk across the portfolio.

In addition, clients can click on any aggregation line from the Aggregated Risk Pane to display the individual trades composing this agg line in the split window below (Constituent Risk Pane).

• Risk Aggregations – Slides

To utilize Risk Slide functionality, user must first create up to 18 customized scenarios. The scenarios can be customized by some combination of price change, vol change, change in days, interest rate change, and change in sDiv.

The screenshot displays the SpiderRock Risk Manager software interface. The main window is titled "Risk Aggregations: All Slides" and shows a table of risk metrics for various assets. A red box highlights the "Risk Aggregations: All Slides" tab. Another red box highlights the "Scenario Customization Window" which is open on the right side of the screen. This window allows users to define 18 scenarios, each with a label, price change (Pct), volatility change (VolChg), and change in days (Chg in Days). The scenarios are listed in a table with columns for #, Label, Pct, VolChg, Chg in Days, and Chg in sDiv.

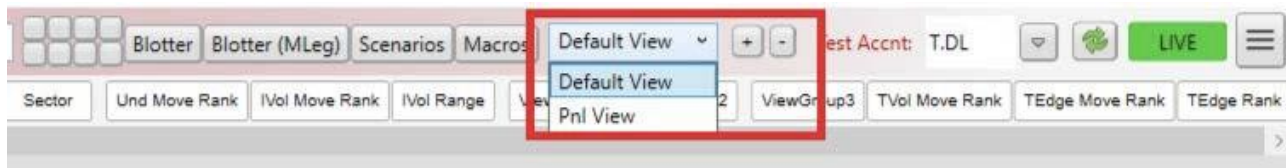
Once scenarios have been created, the client can right-click any of the column headers to select the scenarios they want to incorporate in their slide view. Columns can be arranged in any order, and multiple slide views can be saved.

This screenshot shows the SpiderRock Risk Manager interface with a focus on the "Risk Aggregations" tab. A red box highlights the "Risk Aggregations: All Slides" tab. Another red box highlights the "Scenario Customization Window" which is open on the right side of the screen. This window allows users to define 18 scenarios, each with a label, price change (Pct), volatility change (VolChg), and change in days (Chg in Days). The scenarios are listed in a table with columns for #, Label, Pct, VolChg, Chg in Days, and Chg in sDiv.

- **Creating Risk View**

In addition to the default view, clients can easily create their own custom views by clicking on the + sign to the right of the drop down list which prompts an Add new Aggregation View window. Once the new view is named e.g. PnL View as displayed above, clients can customize their view. Refer to page 8 for column selection details.

Refer to page 5 for information on the Risk Aggregations Slide box.



- **Selecting Aggregation Buckets**

SPIDERROCK Risk Manager

Agg: Accnt Symbol

Clear Accnt SecType Symbol Ticker Key Future Key Option Key Strike Expiration ExpCode Skew EDays Ind Sub Grp Sector

Risk Aggregations Span Aggregations Risk Detail

	count	DayPnL	total.pnL	OpnPnClr	OpnPnLve	\$Delta	Delta	Gamma	Rho	Theta	Vega	WtVega	WVega	CnBot	CnSld	ShBot
	63	(2,075)	249,828	(23,836)	(89,461)	5,693,558	(33,674)	4,592	(311,813)	(34,351)	435,007	74,719	130,951	22	0	400
	Accnt	Symbol	30%up PnL	20%up PnL	15%up PnL	total.pnL	#stk	#opt	sh.bot	sh.sld	cn.bot					
1	T.CH	@ZL	-6,446,755	-4,013,720	-2,799,339	-170,426	0	3	0	0	0					
2	T.CH	AA	1,291	860	645	-38	1	0	0	0	0					
3	T.CH	X	1,116	744	558	36	1	0	0	100	0					
4	T.CH	BABA	906	485	396	-114	0	1	0	0	0					
5	T.CH	F	-602	-401	-301	-21	1	0	0	0	0					

- The Agg selectors (see red arrows), allow clients to aggregate fills using two unique values. If two aggregation values are selected, fills are first aggregated using the value from the first dropdown list and then using the value from the second dropdown list to the right. If the second value is set to 'None', then fills are aggregated using the first value.
- The Totals across the top also follow the same aggregation selection.
- In addition, clients have access to a number of filters across the top where they can further refine their view of risk. For example, a client may be interested to look at a specific symbol or a specific expiration.
- All columns can be sorted by double-clicking the header (double left-click ascending, double leftclick again, descending).

• Additional Filters

Additional filters are available right above the tabs enabling clients to further examine specific areas of their portfolio. Clicking on any filter will bring up a list of values available for a particular filter. Any values can be checked using the square check box to the left. To remove any previously checked filters, clients can click on the funnel in the bottom right of the tool.

SPIDERROCK Risk Manager

Clear Acont SecType Symbol Ticker Key Future Key Option Key Strike Expiration ExpCode Skew EDays Ind Sub Grp Sector

Risk Aggregations Span Aggregations Risk Detail

	count	DayPnL	total.pnl	OpnPnLClr	OpnPnLVe	\$Delta	Delta	Gamma	Rho	Theta	Vega	WtVe
	63	(2,207)	587,063	(27,674)	(4,195)	4,381,573	(34,687)	4,516	(317,080)	(34,445)	433,726	74,9
	Symbol	Symbol	30%up PnL	20%up PnL	15%up PnL	total.pnl	#stk	#				
1	@CL	@CL	-21,030	-14,020	-10,515	-1,090	0					
2	@ES	@ES	12,731,660	8,487,784	6,365,839	-253,763	0					
3	@ZL	@ZL	-6,452,063	-4,020,437	-2,806,562	-146,920	0					
4	AA	AA	1,290	860	645	-41	1					
5	AAPL	AAPL	596,242	383,726	275,289	-7,231	1					
6	AMGN	AMGN	451	104	-67	54	1					

Filter by Ind

- ☐ 0- <none>
- ☐ 1-Basic Materials
- ☐ 2-Conglomerates
- ☐ 3-Consumer Goods
- ☐ 5-Healthcare
- ☐ 6-Industrial Goods
- ☐ 7-Services
- ☐ 8-Technology
- ☐ 9-Utilities

Close All None OK

Furthermore, often used filters can be saved using one of the eight available radio buttons to the left of the blotter box as displayed below. Note, this is in addition to any created custom views.

Blotter Blotter (MLeg) Scenarios Macros Set Binary Default View + - Test Acct: T.CB

Grp Sector Und Move Rank IVol Move Rank IVol Range ViewGroup1 ViewGroup2 ViewGroup3 TVol Move Rank



• Totals

Configurable totals boxes are available and displayed above the grid and below the tabs.

- Right-clicking on any of the total boxes, will open a window of available values that can be checked and added as a total box.
- Totals are responsive to any filters used across the top.

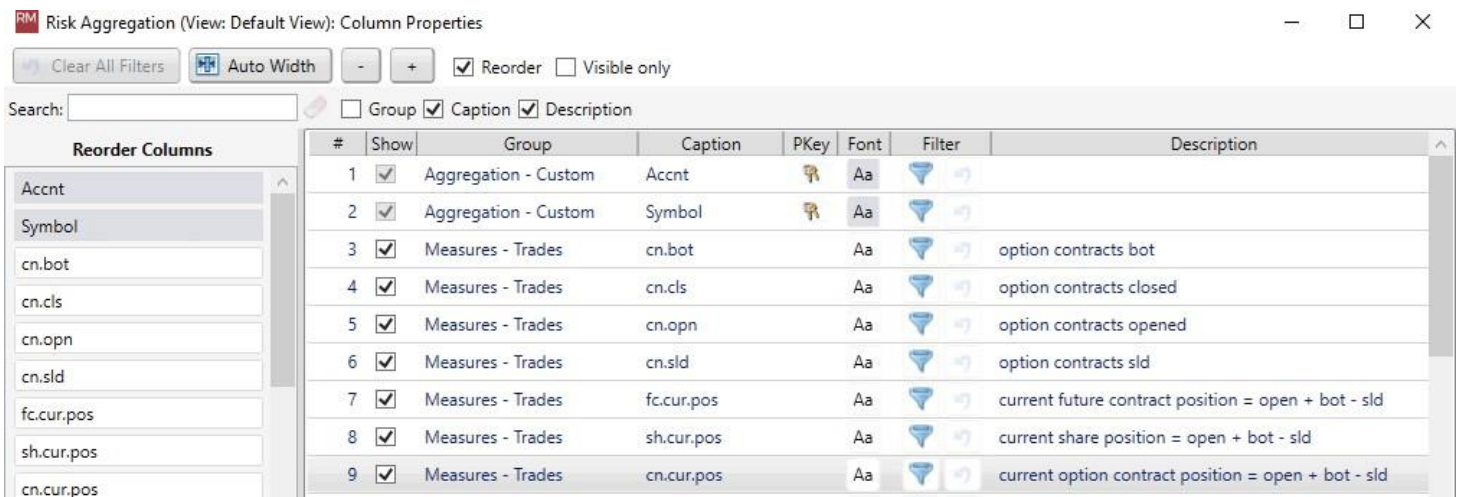
The screenshot shows the SpiderRock RiskManager (v7.0) interface. At the top, there's a row of total boxes for various metrics: Count (63), dayPnL (2,227), totalPnL (636,181), opnPnLClrMark (29,270), opnPnLve (17,713), posDDelta (4,246,509), posDelta (35,142), posGamma (4,572), posRho (319,284), posTheta (34,495), posVega (433,110), posWtVega (74,983), posWVega (131,329), cnBot (22), cnSld (0), shBot (400), shSld (400), and posMktVal (16,122,900). Below this is a grid of symbols (e.g., @CL, @ES, @ZL, AA, AAPL, AMGN, AMZN, BA, BABA, CMTL, CSX, CVS, EBAY, F). A red arrow points to the 'posDelta' total box, which has been right-clicked to open the 'Totals' window.

The 'Totals' window (SpiderRock RiskManager (v7.0) (V7_Stable.Prod | NY4) © 2018 SpiderRock Technology, LLC [c.sampat.2] (ClientRiskTrader) Tot...) contains a search bar and checkboxes for 'Group', 'Caption', 'Description', and 'Show Visible Only'. It lists the following measures:

Sho	Group	Caption	Color	Number Format	Description
☒	Record Count	Count	Aa	#,##0;(#,##0)	Count of number of eligible records
☒	Measures - Trade	dayPnL	Aa	#,##0;(#,##0)	day pnl
☒	Measures - PnL	totalPnL	Aa	#,##0;(#,##0)	OpnPnLVolMark + dayPnL
☒	Measures - PnL	opnPnLClrMark	Aa	#,##0;(#,##0)	(cur.mid - pr.clr) * cnOpnClr * upc
☒	Measures - PnL	opnPnLve	Aa	#,##0;(#,##0)	open position * mult * optVega * dVol
☒	Measures - Live I	posDDelta	Aa	#,##0;(#,##0)	live position * mult * hedgeDe * UPrc
☒	Measures - Live I	posDelta	Aa	#,##0;(#,##0)	live position * mult * hedgeDe
☒	Measures - Live I	posGamma	Aa	#,##0;(#,##0)	live position * mult * optGamma
☒	Measures - Live I	posRho	Aa	#,##0;(#,##0)	live position * mult * optRho
☒	Measures - Live I	posTheta	Aa	#,##0;(#,##0)	live position * mult * optTheta
☒	Measures - Live I	posVega	Aa	#,##0;(#,##0)	live position * mult * optVega

• Column Properties Features

- Columns can be moved left or right by clicking and dragging columns within the Aggregated Risk Pane. To adjust the order of all columns at once, open Column Properties by right clicking on any headers or by clicking on the settings box (top right corner) and then click on Column Properties. In the Column Properties window click the 'Reorder' box which will display all columns in the left panel. Clients can drag and drop all available columns into a specified sequence.
- To change the width of individual columns, left-click the divider between column headers and drag left or right. The width of all columns may be adjusted from the Column Properties menu.



From this window, the user can select columns and filters to display.

- The 'Caption' header shows the name of the column.
- Users can check any values to populate the grid.
- Right-clicking anywhere in this window allows clients to check all or uncheck all values.
- The 'Font' column allows users to set the foreground and background colors for each column.
- The 'Description' column provides specific calculations and further information.
- The 'Search' box allows the user to find a metric without scrolling through the entire list. The search can be done at the 'Group', 'Caption' and 'Description' level by checking the appropriate box to the right of the search window.
- The (+) and (-) buttons allow the user to adjust the width of the columns in the Risk Manager. The Auto Width button will resize columns to the smallest width required to display all data.



- **Creating Risk Scenarios**

Clients can compose up to eighteen distinct risk scenarios by clicking on the Scenarios button displayed below (see red box). These risk scenarios enable clients to shock their portfolios either in price terms, volatility terms, change in expiration dates, change in rates, or percent change in dividend.

Once created, a scenario can apply to any Greeks, as well as PnL, and can be selected using the Column Properties window as described on Page 8.

CH2) © 2021 SpiderRock Technology, LLC [chantelle.sampat] (SRAdmin)

#	Used	Label	PrcChg	ChgType	VolChg	ChgType	Chg in Days	Chg in Rate	Chg in SDiv
Scen01	☒	Drop10	0.00	Pct	0.00	Pct	-5.00	0.00	0.00
Scen02	☒	VOLUP	0.00	Pct	0.00	Pct	0.00	0.00	0.00
Scen03	☒	Scen03	0.00	Pct	0.00	Pct	0.00	0.00	0.00
Scen04	☐	Scen04	0.00	Pct	0.00	Pct	0.00	0.00	0.00
Scen05	☐	Scen05	0.00	Pct	0.00	Pct	0.00	0.00	0.00
Scen06	☐	Scen06	0.00	Pct	0.00	Pct	0.00	0.00	0.00
Scen07	☐	Scen07	0.00	Pct	0.00	Pct	0.00	0.00	0.00
Scen08	☐	Scen08	0.00	Pct	0.00	Pct	0.00	0.00	0.00

- **Creating Market Scans**

Market scans are created from the Macros button as displayed below. Macros were designed for clients to scan the market in real-time based on a comprehensive set of filters to define the stock universe, the option universe, as well as edge criteria if desired. These scans can be used as a trading ideation tool and can also be used to increase or decrease risk across the portfolio.

Please refer to the **Macro Scanner concept guide** for more information.



- **Risk Shaping/Basket Trading Functionality:**

Opening/increasing risk can be done in two ways:

- Market scans (macros) can be used to put on new positions.
- Basket Trading

Clients can launch a Basket Blotter for single and multiple leg orders by clicking on the Blotter or Blotter (MLeg) buttons as displayed below:



A blank blotter can be populated from copying orders from Excel and clicking on the Import button of the Blotter (red rectangle below).

The Excel example below is showing the mandatory fields required to populate a Blotter. There are a number of non-mandatory fields that can be added to create orders. For a list of these fields, please contact us at cs@spiderrock.net.

ticker	src	at	exp	cp	strike	side	sectype
SNAP	NMS	EQT	1/21/2022	C	47	B	o
SPX	NMS	EQT	3/18/2022	P	3850	S	o
MSFT	NMS	EQT	8/20/2021	C	210	B	o
AAPL	NMS	EQT	9/17/2021	C	115	S	o
BA	NMS	EQT	5/21/2021	P	230	B	o

The orders imported from Excel are shown below in the Risk Manager Basket Blotter. Once imported, clients can set algo parameters across the top of the Blotter. See our **Order Entry Concept Guide** for more information.

T.C.B.Blotter-0														
Presets			Order Type/Post Type			Limit Class			Limit Adjuster (value/reference)			Probability Limits		
1 2 3			PostOnly NoPost			Simple Vol			0.00 abs / join =bid =ask			Static MakeLimit: 0.50 TakeLimit: 0.50		
4 5 6			Time In Force Day			Max Visible 100 % all			Num Exchanges 1 Cx/OverLap No PublicSize: None Exchanges (ALL)			OpxFloor 0.00 Size/Adjust Set Sizes Adjust to: 100 %		
												Auto Hedge None Sym Day \$Delta Limit (Buy / Sell) B: 1,000,000 S: 1,000,000 Auto		
												Take Reach/Size Release Immediate Immediate Live		
												UPric Range (%) Min: 2.0 Mac: 2.0 Options Send-At Import Discard		
Acct	SecurityKey	Side	Size	uLimitVol	uLimitPrc	mkEffLimitPx	tkEffLimitPx	Open	Filled	State	mkLimit	tkLimit	tk.surf.woff	refRMetric1Src
1 T.C.B	SNAP-NMS-EQT-2022-01-21-47-C	Buy	0	51.041	12.400	12.400	12.400	0	New				0.000	None
2 T.C.B	SPX-NMS-EQT-2022-03-18-3850-P	Sell	0	22.144	198.900	198.900	198.900	0	New				0.000	None
3 T.C.B	MSFT-NMS-EQT-2021-08-20-210-C	Buy	0	29.912	37.946	37.950	37.950	0	New				0.000	None
4 T.C.B	AAPL-NMS-EQT-2021-09-17-115-C	Sell	0	30.158	15.251	15.250	15.250	0	New				0.000	None
5 T.C.B	BA-NMS-EQT-2021-05-21-230-P	Buy	0	30.635	4.550	4.550	4.550	0	New				0.000	None

• Closing/Decreasing Risk

- Custom market scans can be used to close risk against some/all positions in the portfolio by setting the position type of the macro to 'CloseExact' or 'CloseSmallIRR'.
- Users can also close highlighted positions in the grid by right clicking and selecting 'Close Position'. This will open a window where the user can select 100%, 75%, 50%, 25% (see example below). Users can choose whether to include stock, options, and futures by checking the appropriate box(es). Once selected, clicking OK will launch a blotter with the corresponding orders to close these positions. If user wants to override the size once the Risk Blotter is open, they can do so by adjusting the "Size/Adjust" Factor amount to any percentage.

Risk Aggregations

Span Aggregations

Risk Detail

count

63

DayPnL

(2,061)

total.pnl

398,409

OpnPnlClr

(27,690)

OpnPnlVe

(82,943)

\$Delta

4,676,532

Delt

(34,7

Symbol

▲

Symbol

30%up PnL

2

1

@CL

@CL

-21,009

2

@ES

@ES

12,733,950

3

@ZL

@ZL

-6,467,997

4

AA

AA

1,288

5

AAPL

AAPL

597,314

6

AMGN

AMGN

7

AMZN

AMZN

8

BA

BA

9

BABA

BABA

10

CMTL

CMTL

11

CSX

CSX

12

CVS

CVS

-6,035

Launch Blotter

Close Position

Beta testing

Test

High Probability

Risk Aggregations

Span Aggregations

Risk Detail

count

63

DayPnL

(2,052)

total.pnl

554,763

OpnPnlClr

(27,580)

OpnPnlVe

(64,848)

\$Delta

4,310,081

Delt

(34,7

Symbol

▲

Symbol

30%up PnL

20

1

@CL

@CL

-21,006

2

@ES

@ES

12,733,970

3

@ZL

@ZL

4

AA

AA

5

AAPL

AAPL

6

AMGN

AMGN

7

AMZN

AMZN

8

BA

BA

9

BABA

BABA

10

CMTL

CMTL

11

CSX

CSX

12

CVS

CVS

13

FB

FB

Close What

☒ 100%

☐ 75%

☐ 50%

☐ 25%

☒ Include Options?

☒ Include Stock?

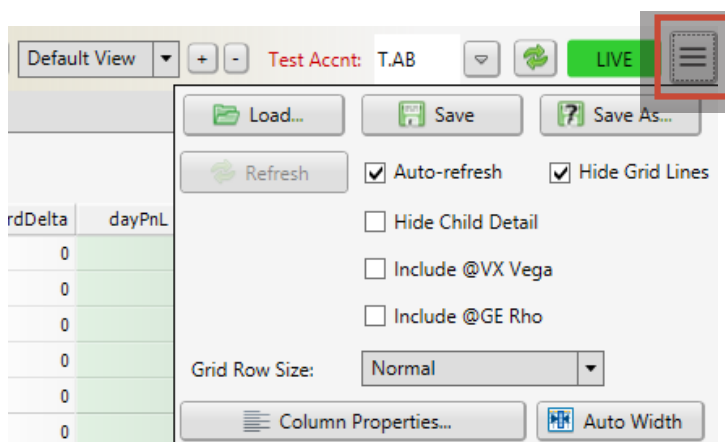
☒ Include Futures?

☒ Cancel

☒ OK

General Menu Functions

Accts:	Gives the user access to permissioned accounts
Views:	Two views are available: Default view and PnL view. Users can create their own views by clicking on the '+' to the right of the view dropdown menu
Manual Refresh:	Allows the user to manually refresh all currently displayed data even when data is Paused
Live/Paused:	This button toggles the window from Live to Paused. When Live, market data and position information flows into the tool in real time. When Paused, the data does not update allowing the user to analyze data in a static view
Settings:	This gives the user access to a variety of general window settings



Load	load saved settings from other users within the same Client Firm that have made their settings Public
Save	save all column settings, including views and filters, for easy retrieval
Auto-refresh	by default, this allows changes to column settings to take effect immediately. If un-checked, changes will not take effect until the user clicks the Refresh button
Hide Grid Lines	by default, grid lines are hidden. If un-checked, the grid lines between boxes will be shown
Hide Child Detail	remove the Constituent Risk Pane
Include @VX Vega	include VIX futures (# of contracts by point value) in Vega calculation
Include @GE Rho	include Euro \$ futures (# of contracts by point value) in Rho calculation
Grid Row Size	allows the user to view in Normal or Compact mode. Compact reduces the row height of all visible rows
Column Properties	allows the user to select the visible columns and filters (see 'Column Properties' section below)
Auto Width	resize columns to the smallest width required to display all data

SPAN AGGREGATIONS TAB

The **Span Aggregations** tab provides a Standard Portfolio Analysis of Risk (SPAN) methodology to evaluate risk under a broad range of 'what-if' market scenarios. The customary 16 slides based on an underlier price and vol change are displayed.

The tool captures all live order activity in real time. All columns can be sorted by double left-clicking the column headings (double left click ascending, double left click again, descending).

OCK Risk Manager

Agg: Acct Symbol Blotter Blotter (MLeg) Scenarios Macros Set Binary Default View Test Acct: T.CB LIVE

Risk Aggregation Source: Custom Prc: 20.00 Pct Master: None SymAdj: None Hedge: DeNeutral Vol Settings: Vol: 0.0 Pct Slide: FixedStrike Days Settings: Adj: 0.0 Max: 0.0 Span Metrics: Price Delta Gamma Theta Vega WtVega

		prc+33%.vol.dn	prc+33%.vol.up	prc+33%.vol.dn	prc+67%.vol.up	prc+67%.vol.dn	prc+67%.vol.up	prc+67%.vol.dn	prc+100%.vol.dn	prc+100%.vol.up	prc+100%.vol.dn	prc+100%.vol.up	prc+300% / 3.0	prc+300% / 3.0	prcspan	prc+100%.vol
1	57	-16,667	-17,261	-17,261	-62,229	-62,229	-67,605	-67,605	-125,074	-125,074	-139,376	-139,376	-186,463	-205,077	0.00	-13
2	50	-460	313	313	-1,087	-1,087	1,015	1,015	-1,432	-1,432	1,408	1,408	-1,665	1,514	0.00	0.00
3	79	-52	-52	-52	-96	-96	-98	-98	-89	-89	-88	-88	-82	-82	0.00	0.00
4	39	539	802	802	1,820	1,820	2,412	2,412	2,884	2,884	3,533	3,533	3,828	3,944	0.00	0.00
5	30	580	313	313	330	330	104	104	-44	-44	-286	-286	-358	-421	0.00	0.00
6	32	3,742	4,745	4,745	10,619	10,619	14,840	14,840	17,794	17,794	25,559	25,559	20,097	29,704	0.20	0.00

- Users can isolate one account or select multiple accounts to run span aggregation risk. See red box on account dropdown menu to the right.
- The Agg selectors (see red arrows), allow the user to aggregate risk by two unique criteria from each of the Agg dropdown menus. If two aggregation criteria are selected, the Risk Manager will first aggregate by the left Agg dropdown menu and then by the right Agg dropdown menu. By selecting 'All' from the right Agg dropdown menu, only a single criteria will be considered for aggregation.
- Users can select any available Span Metrics (Price, Delta, Gamma, Theta, Vega and WtVega) to view span aggregation risk across selected accounts.
- The Span Aggregation tab is split between the Aggregated Risk Pane and the Constituent Risk Pane (see below).

Risk Aggregations Span Aggregations Risk Detail

Source: Custom Price Settings: Prc: 20.00 Pct Master: None SymAdj: None Hedge: DeNeutral Vol Settings: Vol: 30.0 Pct Slide: FixedStrike Days Settings: Adj: 0.0 Max: 0.0 Span Metrics: Price Delta Gamma

	Acct	Symbol	metric	cur.pos	prc.unch.vol.up	prc.unch.vol.dn	prc+33%.vol.up	prc+33%.vol.dn	prc+33%.vol.up	prc+33%.vol.dn	prc+67%.vol.up	prc+67%.vol.dn	prc+67%.vol.up	prc+67%.vol.dn	prc+100%.vol
1			Price	8,427	3,558,623	-3,580,578	3,683,338	797,569	3,190,177	-3,533,336	3,828,982	-3,422,733	2,608,129	-3,297,159	4,2
2			Delta	-34,815	234	552	22,712	31,684	-25,617	-40,898	34,033	41,315	-42,813	-59,126	
3			Gamma	4,575	-987	1,655	-2,613	-3,368	-1,534	-944	-3,876	-4,504	-3,193	-4,480	
4			Vega	433,142	-2,480	1,271	-98	-21,795	-24,280	-33,967	-10,556	-78,479	-62,150	-116,571	
5			Theta	-34,524	-11,601	11,715	-10,159	13,864	-4,319	18,348	-5,896	17,280	3,616	25,407	
6			WtVega	74,959	-347	60	-618	0	-3,402	-15,599	-12,253	-22,074			
7	T.CH	@ZL	Price	-14	-368,994	352,374	-764,580	-438,854	-1,020,694	-485,948	-1,775,387	-1,700,955	-2,671,787	-2,558,544	-3,
8	T.CH	@ZL	Delta	-3	1	-1	-7	-10	11	15	-10	-11	17	19	
9	T.CH	@ZL	Gamma	-6	1	-2	4	5	2	1	5	6	4	6	
10	T.CH	@ZL	Theta	0	0	0	0	0	0	0	0	0	0	0	
11	T.CH	@ZL	Vega	-1	0	0	0	1	0	1	1	1	1	1	

Aggregated Risk Pane

	acct	sec.key	cur.pos	ref.prc	ref.vol	prcspan	d.prc.dn	d.prc.up	volspan	d.vol.dn	d.vol.up	beta	mkt.vol	sym.vol	prc.unch.vol.up	unit.pos.sz	prc.unch.vol.dn
1	T.CH	OZL-CBOT-FUT-2018-09-21-28-C	0.09	28.5682	0.1631	0.200	5.7136	5.7136	0.300	0.0489	0.0489	1.00	1.000	0.171	0	-10	0
2	T.CH	OZL-CBOT-FUT-2018-09-21-28.5-C	0.11	28.5682	0.1667	0.200	5.7136	5.7136	0.0000	0.0000	0.0500	1.00	1.000	0.171	0	-11	0
3	T.CH	ZL-CBOT-FUT-2018-08-14	0.00	28.3100	0.0000	0.200	5.6620	5.6620	0.0000	0.0000	0.0000	1.00	1.000	0.171	0	17	0
4	T.CH	OZL-CBOT-FUT-2018-09-21-27.5-C	0.08	28.5682	0.1601	0.200	5.7136	5.7136	0.300	0.0480	0.0480	1.00	1.000	0.171	0	-10	0

Constituent Risk Pane

Symbol	metric	cur.pos	prc.unch.vol.up	prc.unch.vol.dn	prc+33%.vol.up	prc-33%.vol.dn

- **Source Settings**

Users have the choice between a default Exchange mode and a Custom mode. When Exchange is selected, all price and vol settings will default to set values (CME Group). When Custom is selected, the following price and vol settings are available (see below).

- **Price Settings**

Prc input: Numeric price shock of the reference index/ETF

Prc dropdown: The price move of the reference index/ETF can be expressed as a Percent, Log, or in Points

Master: Reference index/ETF: SPX, IWM, QQQ, VIX, USR (user-defined beta), None

SymAdj: Normalization metric based on Beta, VolRatio, nVolRatio, rSymVol_X1, rSymVol_X2, rSymVol_X3, None

Hedge: Risk-adjusted metric: UseExisting, DeNeutral, AdjDeNeutral, NoHedge

note: Users can upload their own betas per expiration using SRSE analytics (see **SRSE concept guide** for more information). These user-defined betas will be displayed in the lower grid of the Span Aggregation tab.

- **Vol Settings**

Vol input: Numeric vol shock of the reference index/ETF

Vol dropdown: Percent, Log, vPts, Absolute move of the reference index/ETF, or Annualized Vol

Slide: Static (FixedStrike) or dynamic (SlideSurf) slide

- **Days Settings**

Adj: If clicked, span aggregation risk will be displayed for the length of time in days specified, ahead from present.

Max: Users have the ability to set a maximum number of days to look at span aggregation risk. This is often used in the situation of a takeover.

- **Span Metrics**

The span calculation can apply to any of the following metrics: Price, Delta, Gamma, Theta, Vega and WtVega.

Options					
Gamma ☒ Theta ☒ Vega ☒ WtVega				Recalc	
On	Prc-100%.VolUp	Prc-100%.VolDn	Prc +300% /3.0	Prc -300% /3.0	maxLoss
,985	-2,592,307	-2,593,027	-6,651,299	-21,374	-6,651,299
,621	362,331	362,138	-849,858	1,741	0
,568	78,603	78,589	78,395	-195	0
,148	11,768	11,680	11,569	-709	0
,370	-35,973	-35,889	-38,335	165	0
,009	21,938	21,922	22,723	-59	0
8,072	-29,997	-29,997	-8,772	-36,788	-36,788
-926	4,074	4,074	-926	4,074	0
603	603	603	603	603	0

- Worst possible loss values are clearly identified within each scenario in the red cells above
- The maxLoss column reflects the worst possible scenario from all 16 shocks
- Additional filters are available at the top of Risk Manager to target risk with more granularity

RISK DETAIL TAB

The Risk Detail tab contains a single row for every strike and underlying position in the selected account in a non-filtered capacity. Users can easily export the content to Excel for further analysis. To highlight the entire grid, click on the box highlighted in green below. Then hit Ctrl + C to copy and Ctrl + V to paste into Excel.

SPIDERROCK Risk Manager

Clear

Accont

SecType

Symbol

Ticker Key

Future Key

Option Key

Strike

Expiration

ExpCode

Skew

EDays

Risk Aggregations

Span Aggregations

Risk Detail

root

sec.type

expiration

strike

opn.pnl.vol.mark

opn.pnl.ga

opn.pnl.de

opn.pnl.th

opn.pnl.ve

1

FB

Stock

2000-00-00

0.00

-6,064

0

-6,046

0

2

F

Stock

2000-00-00

0.00

-23

0

-23

0

3

EBAY

Stock

2000-00-00

0.00

58

0

46

0

4

CVS

Stock

2000-00-00

0.00

-172

0

-154

0

5

CSX

Stock

2000-00-00

0.00

-42

0

-42

0

6

CMTL

Put

2018-10-19

20.00

-1

0

0

0

-

7

CMTL

Put

2018-10-19

25.00

6

0

1

-1